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Automatic Door & Gate Drive Fault-Finding Checklist

For facility engineers, gate installers and service technicians working on sliding doors, sliding gates and swing gates

Work through the steps in order: they move from the quickest, safest checks to those that need tools and isolation. Record every reading, and compare it with the commissioning values kept with the operator, so each diagnosis is a comparison rather than a guess.

Step 1 — Safety and isolation

- [ ] Isolate the mains supply and disconnect the back-up battery before working inside the operator or on the track; lock out and tag the isolator.
- [ ] Keep hands clear of rack, pinion, belt pulleys and hinge pinch points; a released gate can move under wind or slope.
- [ ] Support heavy sliding leaves and swing leaves before removing wheels, hinges or the operator.
- [ ] Confirm that photocells, safety edges and stop buttons work before returning the door to automatic mode.

Step 2 — Controller and safety inputs

- [ ] Read and record the controller fault code or status LED pattern before resetting anything.
- [ ] Check each photocell pair: clean the lenses, confirm the alignment LED, and look for low sun or reflective surfaces in the beam path.
- [ ] Test each safety edge and stop button: the door must stop or reverse when it is activated.
- [ ] Watch whether the radar or motion sensor triggers with nobody present (door reopens by itself).
- [ ] Note the force or sensitivity setting; do not raise it until friction faults are cleared.

Step 3 — Door, track and wind (manual-release push test)

- [ ] Engage the manual release and move the leaf through full travel by hand; it should move evenly with no tight spots.
- [ ] Measure push force with a push-pull gauge or spring balance; investigate any value above 1.5 × the commissioning figure.
- [ ] Inspect wheels, guide rollers, bearings and hinges for flat spots, corrosion and dry bearings; clear grit and stones from the track.
- [ ] Check rack-to-pinion vertical clearance along the full travel (typically 1–2 mm) so the gate weight stays on its wheels.
- [ ] For solid leaves in exposed sites, estimate wind force: $F \approx 0.613 \times v^2 \times A \times Cd$ newtons (Cd about 1.2 for a solid panel).

Step 4 — Motor, gearbox, brake and lock

- [ ] With the leaf disconnected, run the drive and record no-load current and terminal voltage; compare with the test sheet or commissioning record.
- [ ] Measure backlash at the pinion or output shaft; bronze dust or bright metal indicates worm-wheel wear.
- [ ] Measure lock or brake coil voltage during the open command and confirm full release; check the brake air gap against its data sheet.
- [ ] Check brushes against the minimum-length mark and confirm even spring pressure; inspect the commutator for burning or grooving.

- [ ] Measure winding insulation resistance with a 500 V megger after disconnecting all electronics; trend it against earlier readings.

Step 5 — Supply, battery back-up and duty

- [ ] Measure each 12 V back-up battery at rest (about 12.7 V or more when fully charged) and during a door cycle; replace weak blocks as a matched set.
- [ ] Measure motor terminal voltage while the door moves and compare with the supply voltage to find cable or connector drop.
- [ ] Count cycles per hour at the busiest time of day and time each opening and closing movement.
- [ ] Duty factor (%) = motor-on time ÷ total time × 100; compare with the S3 rating on the motor nameplate.

Step 6 — Recommissioning

- [ ] Run the full learn or set-up cycle without interruption after any mechanical repair or part replacement.
- [ ] Record manual push force, running current in each direction and full-travel time; keep them with the operator.
- [ ] Verify that the door stops and reverses on contact with a test obstruction at several points of travel.
- [ ] Log the date, readings and parts replaced so the next service visit starts from a baseline.

Symptom → first check → likely cause

Symptom	First check	Likely cause
Reverses mid-travel, nothing in the way	Push test and running current	Track or roller friction, wind load
Stops short or hits the end stop	Limit hardware, slip marks	Moved limits, lost encoder count, belt slip
Opens but will not close	Fault code, photocell LEDs	Blocked or misaligned photocell, safety edge
Motor hums, leaf does not move	Lock or brake coil voltage	Lock or brake not released, jammed gearbox
Slow or stalls in power cuts	Battery voltage under load	Aged or discharged back-up batteries
Clunk on starting and reversing	Backlash at the pinion	Worn worm wheel or pinion, rack clearance lost
Fails only at peak hours	Cycles per hour, frame temperature	Duty rating exceeded, thermal trip
Intermittent starting, sparking	Brush length, rotate and retry	Worn brushes, commutator dead spot

Values are typical guidance. Always follow the door or gate manufacturer's instructions and the applicable door-safety requirements for force limits and safety devices. For drive selection support, contact Globe Scott Motors through www.globescott.net/contact-us/.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.